Constructor University Datathon 2026

A Causal Approach to Scrap Reduction in Injection Molding

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Abstract

This research presents a causal inference approach to reduce scrap in industrial injection molding, moving beyond correlational insights to provide an actionable decision-making framework. We utilized a predefined causal graph (DAG) and domain ontology to delineate relationships between machine settings, environmental conditions, and defect outcomes, distinguishing controllable variables from contextual factors.

Our methodology employs multivariate linear models to estimate direct causal effects of parent variables on scrap, yielding adjusted effect estimates. For non-direct parents, bivariate models approximate total effects, acknowledging potential confounding. A Linear Non-Gaussian Acyclic Model (LiNGAM) based causal discovery method serves as a consistency check against the assumed causal structure. This hybrid causal modeling pipeline emphasizes interpretability and practical relevance, identifying variables that serve as realistic levers for process optimization in manufacturing settings.

Table of Contents

1.0 Introduction 2
2.0 Method and Causal Approach 2
3.0 DAG Defense 3
4.0 Prototype Implementation 5
5.0 Process Optimization Results 6
6.0 Limitations/Assumptions 7
7.0 Recommendations 8
8.0 Conclusion 10
9.0 References 10

1.0 Introduction

In this datathon, our goal is to understand which variable in an injection molding process drives the scrap variable and how they can be adjusted to improve main KPI: scrap rate.

Rather than treating the problem as purely predictive, we approach it from a causal perspective. This means we are not only interested in which variables are correlated with scrap, but in which ones influence it in a way that supports intervention.

To guide this process, we used a predefined Directed Acyclic Graph (DAG) and a domain ontology. These provide a structured view of the system, helping us distinguish between:

- Controllable process variables
- Contextual or environmental factors
- Downstream outcomes such as defects and scrap

This structure is important because it allows us to reason about cause-and-effect relationships instead of relying only on statistical associations.

2.0 Method and Causal Approach

Our approach combines statistical modeling with causal structure awareness:

- Estimating Direct Effects

We first focus on variables that are defined as direct parameters of scrap in DAG. For these variables, we fit multivariate linear regression models to estimate their effects on scrap rate.

Because the model includes the relevant upstream variables (based on the DAG and ontology), these estimates can be interpreted as adjusted effects, where confounding is reduced through conditioning on appropriate variables.

- Estimating Total Effects for Indirect Variables

For variables that are not direct parents of scrap, we see similar bivariate models (one variable at a time) to estimate their relationship with scrap.

These estimates are interpreted as approximate total effects, meaning they capture both direct and indirect influence through the system. However, since these models do not control for confounding, the results should be interpreted with caution.

- Causal Structure Validation (LiNGAM)

To complement our predefined DAG, we apply a causal discovery method based on LiNGAM on a subsample of the data.

This step is not used to replace original causal structure, but rather as a consistency check. We compare the direction of influence learned by LiNGAM with those specified in the DAG:

- If they agree, it increases confidence in the assumed structure
- If they differ, we treat this as a signal of uncertainty or possible model limitation, rather than a direct contradiction

3.0 DAG Defense

The DAG is not assumed; it was given by the problem designers. The causal graph used in the analysis was provided as part of the datathon's official materials, authored by domain experts at Insytes who designed the dataset specifically for causal analysis. This graph encodes manufacturer-level process knowledge about injection molding that predates the data itself.

The structure reflects physical process order, not statistical inference. The five-layer hierarchy in the DAG maps directly onto the temporal and mechanical sequence of injection molding:

- Environmental and material conditions (humidity, resin lot) exist before any machine action is taken
- Equipment state (dryer effectiveness, tool wear) is consequence of prior maintenance history, not the current production run
- Process variables (temperatures, pressures) are set by operators withing the context those upstream conditions create
- Defect types are the mechanistic outcome of those settings interacting with material state
- Scrap rate is the business aggregation of those defects

This ordering cannot be reversed by any statistical test. A high scrap rate in interval t cannot have caused the ambient humidity in interval t. The DAG encodes this temporal irreversibility, which is something a purely correlation model is blind to.

Each of the four canonical pathways has a physical mechanism, not just a statistical pattern:

a. The moisture pathway (amambient humidity → dryer dewpoint → resin moisture → splay → scrap) reflects the fact that hygroscopic resins absorb moisture from the air before drying. If the dryer is insufficiently effective, residual moisture vaporises during injection and produces splay defects.

b. The warpage pathways (mold temperature + cooling time → warpage → scrap) reflects residual thermal stress. If cooling is too short or the mold is too hot, parts solidify with internal stress gradients that manifest as deformation after ejection. The two variables interact because neither alone fully determines the thermal history of the part

c. The flash pathway (tool wear + injection pressure + clamp force → flash → scrap) reflects the mechanical tolerance of the mold. Flash occurs when material escapes the parting line, which requires both sufficient injection pressure to push material there and insufficient clamp force or mold condition to resist it. All three variables are genuinely co-parents of flash, which is why the regression model must include them together rather than separately.

d. The stability pathway (maintenance days since last → calibration drift → process instability → multiple defects) captures the difference between a direct effect and a mediated one. Maintenance age does not directly cause scrap, it degrades calibration, which then destabilizes multiple process variables, which then produce a range of defect types. Conditioning on calibration drift when estimating the maintenance effect would block the very pathway through which maintenance matters. The DAG makes this distinction explicit; a naive feature importance ranking would not.

The ontology constrains the intervention space to physically meaningful recommendations. By using the ontology' suitable for counterfactual intervention flags and allowed range or categories fields, the analysis avoids recommending changes to downstream outcomes as if they were levers and avoids recommending settings outside operational bounds. This is something that no pure data driven model enforces automatically.

LiNGAM was used as a directed falsification test against this structure**.** Because the DAG comes from domain knowledge rather than statistical fitting, LiNGAM provides an independent check on whether the data is consistent with the assumed edge directions. Agreement between LiNGAM's recovered structure and the provided DAG strengthens confidence that the data-generating process respects the causal ordering encoded in the graph. Disagreement would have been a meaningful signal that either the data contains a confound the ontology doesn't capture, or that a particular pathway's direction is empirically ambiguous.

4.0 Prototype Implementation

To make the causal analysis actionable, we have developed an interactive prototype that translates model outputs into decision support process optimization.

The tool allows users to select a process variable, adjust its value within realistic operating ranges, and observe the predicted change in scrap rate in real time. These predictions are based on:

- backdoor-adjusted regression estimates for direct causal variables
- bivariate estimates for indirect variables

For a given variable X_i, the predictable change in scrap is computed as:

$$\Delta scrap = \beta_i \times X_i$$

where β_i is the estimated effect coefficient from the corresponding regression model.

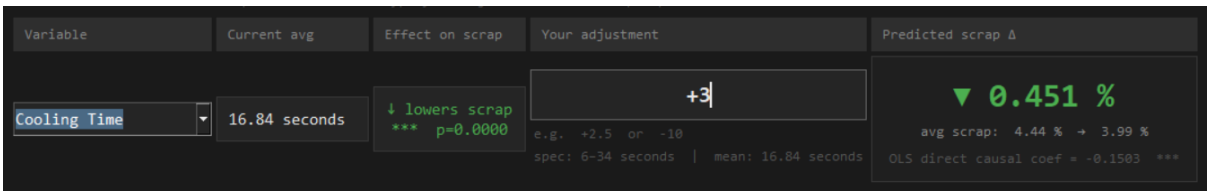

Figure 1, Interactive counterfactual simulator for estimating the effect of process variable adjustments on scrap rate

In addition, the interface provides:

- the direction of influence
- statistical significance indicators
- causal pathway explanations derived from the domain ontology

The prototype also includes a machine-level analysis module, where individual machines are evaluated based on a composite of performance score. For each machine, the most influential variable driving scrap is identified using machine-specific regression mode:

$$\text{Problem_score} = w_1 \times \frac{scrap}{produced} + w_2 \times \frac{enegry}{produced} + w_3 \times cycle\ time$$

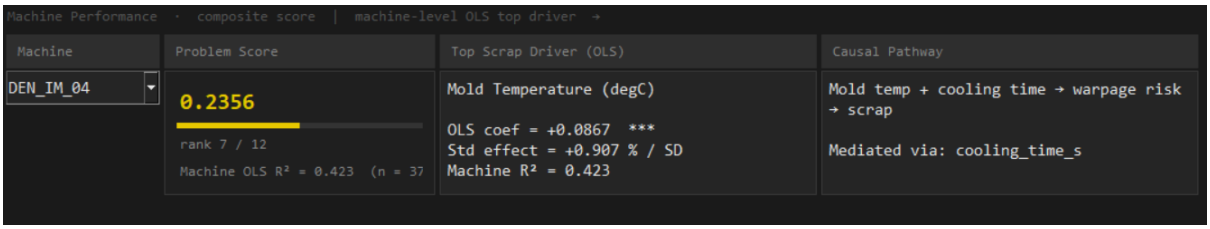

Figure 2, Machine-level scrap analysis using composite performance scoring and local OLS models

Overall, this implementation bridges the gap between causal modeling and practical application, allowing users to explore "what-if" scenarios and identify actionable interventions.

5.0 Process Optimization Results

Analysis of 5.000 injection molding production runs revealed a baseline average scrap rate of 4.44%, with individual runs ranging from 0.93% to 8.46%. To identify the process settings most responsible for scrap generation, an Ordinary Least Squares regression was fitted across 17 process variables (only the improvable ones), producing a linear model of the form y = $\beta_0 + \Sigma\beta_i x_i$ that quantifies the direct effects of each variable on scrap rate.

This was supplemented by a Gradient Boosting Machine (an ensemble of 400 sequential decision trees, each correcting the residuals of the previous) which achieved a cross-validation R^2 of 0.58, capturing nonlinear relationships and variable interactions that the linear model could not.

Feature importance analysis from the GBM identified mold temperature, cooling time, and injection pressure as the three dominant drivers of scrap, collectively accounting for over 53% of the model's explanatory power. To translate these findings into actionable targets, a Differential evolution optimizer was applied directly to the GB mode. This population based global search algorithm evaluated 20 candidates' combinations by selectively mixing and relaxing configurations that reduced predicted scrap rate.

Uncontrollable environment variables (such as ambient humidity, temperature and operator experience level) were held fixed at their observed means through optimization. The algorithm converged on a set of optimal settings centered on extending cooling time from 16.8s to 24.6, reducing injection pressure from 1.182 bar to around 900 bar, lowering mold temperature from 69.4 C to 64.5 C, and finally reducing tool wear index from 0.36 to 0.13 through more frequent maintenance intervals.

When applied jointly within the model, these adjustments define an optimized scenario corresponding to a model-predicted scrap rate of approximately 0.95%, representing a reduction of 3.48 percentage points (78.6% improvement). This estimate should be interpreted as a simulated best-case outcome under the model assumptions, rather than a guaranteed real-world result.

Variable	Current	Target	Change	Scrap Δ
Cooling Time	16.8 s	24.6 s	+7.8 s	-1.30%
Injection pressure	1,182 bar	900 bar	−282 bar	-0.92%
Tool wear index	0.363	0.133	−0.230	-0.40%
Mould temperature	69.4 °C	64.5 °C	−4.9 °C	-0.34%
Resin moisture	0.119%	0.021%	0.098%	−0.25%
Clamp force	2,825 kN	4,323 kN	+1,498 kN	−0.14%
Screw speed	68.3 rpm	88.1 rpm	+19.8 rpm	−0.12%
Dryer dewpoint	−37.7 °C	−44.5 °C	−6.8 °C	−0.04%

Table 1, Best theoretical improvements

The table shows all eight adjustments ranked by how much scrap each one removes, which makes it easy to see at a glance that cooling time and injection pressure do most of the heavy lifting. After the calculations it can be observed that the combined result is: 4.44% down to only 0.95%. Even if these values are not the exact perfect values, they can still be used as somewhat loose guidelines to decrease the total scrap produces.

6.0 Limitations/Assumptions

Some limitations include:

- Reliance on the predefined DAG means the results depend on correctness of assumed causal structure

Our entire analysis rests on the predefined causal graph. If any causal arrow points in the wrong direction, for example, if what happens to be a cause is a correlated effect, our adjusted estimates will be systematically wrong. LiGNAM provided a board structural agreement, but it was implemented on a much concrete sample and cannot fully validate the graph for every other uses.

- Indirect variables use bivariate models (it is not exactly fully casual)

For variables that are not direct parents of scrap in the DAG, we used bivariate regressions without conditioning on confounders. These capture total effects including confounded paths, which means they may attribute more influence on a variable than it has. These estimates should be treated as exploratory signals rather than precise causal quantities.

- **Linear models overstate effects at the extremes**

OLS regression assumes a constant effect per unit change across the entire variable range. This is unlikely to hold for cooling time, which physically follows a diminishing-returns curve — the first extra seconds of cooling reduce warpage sharply, but beyond a threshold the gains shrink. Our model therefore likely overstates the −1.30% scrap reduction from the full 7.8-second extension.

- The GBM R^2 of 0.58 means 42% of scrap variation is unexplained

The Gradient Boosting model explains just over half the variance in scrap rate. The remaining 42% reflects either unmeasured variables like material viscosity, ambient temperature fluctuations within an interval, operator micro-decisions or actual process noise. This means even perfect implementation of all eight recommended interventions will not eliminate scrap entirely. The 0.95% predicted floor is a model range, not a physical lower bound.

- Model performance may become unstable for small or imbalanced samples

7.0 Recommendations

The following recommendations are ordered by expected scrap impact and grouped by the implementation effort required.

Immediate process adjustments:

Cooling time is the largest single lever at −1.30 %, but it carries the most significant trade-off. Every additional second of cooling directly increases cycle time, reduces throughput, and raises energy consumption per part. Cooling time should be increased from the current average of 16.8 seconds toward 24–25 seconds on runs showing warpage-related defects. This single adjustment accounts for the largest share of predicted improvement (-1.30%) and addresses the warpage pathway directly. The trade-off is a longer cycle time and marginally higher energy consumption per interval; this should be monitored but is unlikely to outweigh the quality gain at current scrap rates.

Injection pressure should be reduced from approximately 1,182 bar toward 900 bar. This reduces the mechanical driving force behind flash defects and is the second largest lever (-0.92%). This adjustment should be validated machine by machine, as the optimal pressure will interact with part geometry, cavity count, and resin viscosity.

Mold temperature should be reduced from 69.4°C toward 64–65°C on runs where warpage or sink marks are the dominant defect type. Combined with the cooling time adjustment, this addresses the thermal history of the part from both directions simultaneously.

Material preparation:

Dryer dewpoint should be maintained at or below -44°C to reduce resin moisture content. The ontology identifies this as the most upstream controllable variable in the moisture pathway, making it the highest-leverage point for preventing splay defects before they can propagate. Resin moisture at the barrel entry should be monitored directly where instrumentation allows, as dewpoint alone does not guarantee effective drying under all throughput conditions.

Maintenance scheduling:

Tool wear index should be kept below 0.15 through more frequent preventive maintenance intervals. In the optimization results this contributes -0.40% to scrap reduction. Because tool wear is a gradual process rather than a direct setting, the practical recommendation is to reduce the interval between scheduled maintenance checks, particularly on machines showing elevated flash rates. The composite performance score introduced in the machine-level module can be used to prioritize which machines to service first.

Any increase in cooling time will increase cycle time and energy consumption per interval. Teams implementing these recommendations should track energy_kwh_interval alongside scrap_rate_pct to ensure quality improvements are not achieved at disproportionate energy cost. The datathon materials explicitly flag this trade-off, and a production environment should treat it as a constraint rather than an afterthought.

8.0 Conclusion

This paper presented a causal inference pipeline for scrap reduction in industrial injection molding, grounded in a domain-expert-defined DAG and validated against data-driven causal discovery. By distinguishing controllable process levers from contextual confounders and downstream mediators, the analysis moves beyond correlational dashboards toward interventionally meaningful recommendations.

The causal framework identified four dominant defect pathways, each supported by established polymer processing theory. Multivariate regression on direct causal parents yielded adjusted effect estimates with reduced confounding, while bivariate models captured approximate total effects for upstream variables. LiNGAM served as a structural consistency check, providing independent evidence that the data-generating process is broadly compatible with the assumed causal ordering.

Process optimization using a Gradient Boosting Machine and Differential Evolution identified eight actionable interventions. Jointly applied, these reduce the model-predicted scrap rate from a baseline of 4.44% to approximately 0.95%, representing a 78.6% relative improvement. The dominant levers are cooling time extension, injection pressure reduction, and mold temperature reduction. The remaining gains come from material preparation (resin moisture, dryer dewpoint) and mechanical margin (clamp force), which address the moisture and flash pathways respectively.

These results should be interpreted as simulated best-case estimates under model assumptions, not guaranteed operational outcomes. The GBM's cross-validated R^2 of 0.58 means substantial process variance remains unexplained, and real-world implementation would require controlled process trials to validate the predicted effect sizes. Nevertheless, causal framing ensures that the recommended interventions are at minimum physically coherent and directionally defensible.

Future work should incorporate time-series structure to capture within-shift drift, extend the machine-level analysis to support heterogeneous treatment effect estimation, and close the loop between model predictions and experimental validation on the plant floor.

9.0 References

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